

and technology. However, the technology is still in its early stages of development, and there are significant challenges to be addressed in terms of hardware limitations, error correction, and algorithm development, as well as scalability and reliability. Nonetheless, with continued research and development, quantum computing could lead to significant breakthroughs and advancements in many areas of science and technology.

In 2020, Google claimed to have achieved quantum supremacy with its Sycamore quantum computer, which took just 200 seconds to complete a calculation that would have taken the world's fastest supercomputer 10,000 years to complete.

Other major players in the quantum computing field include IBM, Microsoft, Intel, and Honeywell, all of whom are developing their own quantum computing technologies and making them available to researchers and developers through cloud based services.

Key advantages of quantum computing

Speed: Quantum computers can solve certain problems much faster than classical computers, especially those related to data optimisation, machine learning, and cryptography.

Parallelism: Quantum computing allows for massive parallelism, which means that many calculations can be performed at the same time.

Quantum superposition: Quantum computing can make use of quantum superposition, which allows quantum bits (qubits) to exist in multiple states simultaneously. This allows for more complex calculations and faster problem-solving.

Quantum entanglement: Quantum computing can also make use of quantum entanglement, which allows qubits to be connected in such a way that the state of one qubit affects the state of the others. This can be used to perform certain types of calculations much faster than classical computing.

Improved accuracy: Quantum computing can offer improved accuracy over classical computing in certain calculations, such as in the simulation of chemical reactions and the modelling of financial markets.

Security: Quantum computing can potentially offer improved security over classical computing in areas such as cryptography, as certain algorithms that are difficult to break using classical computing can be easily broken by quantum computing.

Innovative applications: Quantum computing is a new and rapidly evolving field, with the potential for a wide range of innovative applications in areas such as drug discovery, materials science, and artificial intelligence.

Some of the potential applications of quantum computing are listed below.

- **Cryptography:** Quantum computers have the potential to break many of the cryptographic algorithms that are used

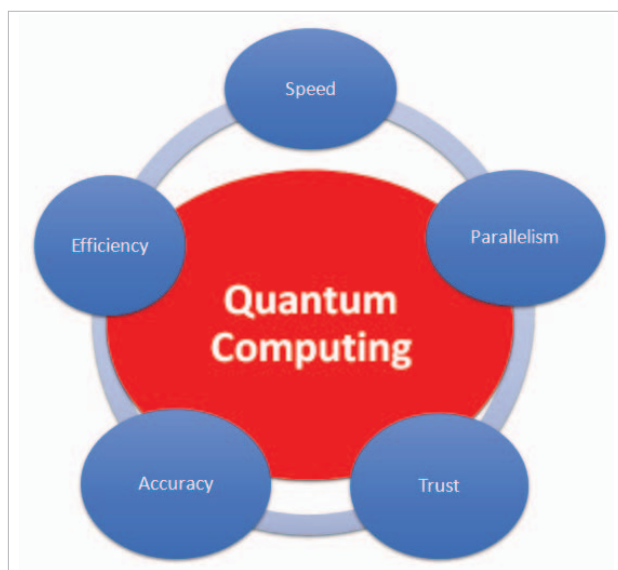


Figure 1: Advantages of quantum computing

to secure sensitive data. However, they can also be used to develop new, more secure encryption methods.

- **Optimisation:** Many real-world optimisation problems, such as supply chain management and logistics, are extremely difficult to solve with classical computers. Quantum computers can provide faster and more efficient solutions to these problems.
- **Machine learning:** Quantum machine learning algorithms can be used to analyse and classify large amounts of data more efficiently than classical algorithms.
- **Chemistry:** Quantum computers can simulate the behaviour of molecules and chemical reactions more accurately than classical computers, which can lead to the development of new materials and drugs.
- **Finance:** Quantum computing can be used to optimise portfolios, risk assessment, and other financial calculations.
- **Weather forecasting:** Quantum computing can provide more accurate and precise weather forecasts by simulating complex weather patterns and climate models.
- **Particle physics:** Quantum computing can be used to simulate particle interactions, and accelerate the development of new theories and technologies in particle physics.

Machine learning integration with quantum computing

Quantum machine learning (QML) is a field that explores the intersection between quantum computing and machine learning. It is focused on developing algorithms and techniques that can leverage the unique properties of quantum computing to improve the efficiency and accuracy of machine learning tasks.